

For the planar velocity field

$$v_r = -A \sin(\Omega t) \frac{\sin \theta}{r^2}$$

$$v_\theta = A \sin(\Omega t) \frac{\cos \theta}{r^2},$$

where A and Ω are known constants,

1. Determine the vorticity.
2. Compute the expansion rate.
3. Comment on the existence of velocity potential and stream function and, if they do exist, compute them.
4. Find the acceleration for points along the line $\theta = 0$
5. Obtain the trajectories & streamlines
6. Determine the fluid line that at $t = 0$ is given by $\theta = 0$ and $0 < r < \infty$.

$$\textcircled{1} \quad \omega_z = \frac{1}{r} \left(\frac{\partial(r v_\theta)}{\partial r} - \frac{\partial v_r}{\partial \theta} \right) = \frac{1}{r} \left(-A \sin(\Omega t) \frac{\cos \theta}{r^2} + A \sin(\Omega t) \frac{\cos \theta}{r^2} \right) = 0$$

$$\textcircled{2} \quad \nabla \cdot \vec{v} = \frac{1}{r} \left(\frac{\partial(r v_r)}{\partial r} + \frac{\partial v_\theta}{\partial \theta} \right) = \frac{1}{r} \left(A \sin \Omega t \frac{\sin \theta}{r^2} - A \sin(\Omega t) \frac{\sin \theta}{r^2} \right) = 0$$

$\textcircled{3}$ THERE EXISTS VELOCITY POTENTIAL BECAUSE $\nabla \wedge \vec{v} = 0$

$$\vec{v} = \nabla \psi \rightarrow v_r = \frac{\partial \psi}{\partial r} = -A \sin(\Omega t) \frac{\sin \theta}{r^2} \Rightarrow \psi = A \sin(\Omega t) \frac{\sin \theta}{r} + f(\theta)$$

$$v_\theta = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = A \sin(\Omega t) \frac{\cos \theta}{r^2} \Rightarrow \frac{1}{r} \frac{\partial \psi}{\partial \theta} = A \sin(\Omega t) \frac{\cos \theta}{r^2} + \frac{f'(\theta)}{r} = A \sin(\Omega t) \frac{\cos \theta}{r^2}$$

$$\downarrow \\ f'(\theta) = 0 \rightarrow f = \text{const} = \psi_0$$

$$r v_r = \frac{\partial \psi}{\partial \theta} = -A \sin(\Omega t) \frac{\sin \theta}{r} \Rightarrow \psi = A \sin(\Omega t) \frac{\cos \theta}{r} + g(r)$$

$$v_\theta = -\frac{\partial \psi}{\partial r} = A \sin(\Omega t) \frac{\cos \theta}{r^2} \Rightarrow A \sin(\Omega t) \frac{\cos \theta}{r^2} - g' = A \sin(\Omega t) \frac{\cos \theta}{r^2} \Rightarrow g' = 0 \rightarrow \psi = A \sin(\Omega t) \frac{\cos \theta}{r} + \psi_0$$

$$\boxed{\psi = A \sin(\Omega t) \frac{\sin \theta}{r} + \psi_0}$$

$$\textcircled{4} \quad \vec{a} = \frac{d\vec{v}}{dt} + \vec{v} \cdot \nabla \vec{v} = \frac{d\vec{v}}{dt} + \nabla \left(\frac{v^2}{2} \right) - \vec{v} \wedge (\nabla \wedge \vec{v})$$

$$a_r = \frac{dv_r}{dt} + \frac{\partial}{\partial r} \left(\frac{v^2}{2} \right) = -A \Omega \cos(\Omega t) \frac{\sin \theta}{r^2} - 2A^2 \frac{\sin^2(\Omega t)}{r^5}$$

$$a_\theta = \frac{dv_\theta}{dt} + \frac{1}{r} \frac{\partial}{\partial \theta} \left(\frac{v^2}{2} \right) = A \Omega \cos(\Omega t) \frac{\cos \theta}{r^2} - 2A^2 \frac{\sin^2(\Omega t)}{r^5}$$

$$\boxed{\frac{v^2}{2} = \frac{A^2 \sin^2(\Omega t)}{2 r^4}}$$

$$\theta = 0 \rightarrow \begin{cases} a_r = -2A^2 \frac{\sin^2(\Omega t)}{r^5} \\ a_\theta = A \Omega \frac{\cos(\Omega t)}{r^2} \end{cases}$$

$\textcircled{5}$ TRAJECTORIES:

$$\frac{dr}{dt} = v_r = -A \sin(\Omega t) \frac{\sin \theta}{r^2} \quad \frac{r d\theta}{dt} = v_\theta = A \sin(\Omega t) \frac{\cos \theta}{r^2}$$

$$t=0: r=r_0, \theta=\theta_0$$

$$\frac{dr}{r d\theta} = -\frac{\sin \theta}{\cos \theta} \rightarrow \frac{r}{r_0} = \frac{\cos \theta}{\cos \theta_0}$$

THESE ARE ALSO THE STREAMLINES

$$\frac{d\theta}{dt} = A \sin(\Omega t) \frac{\cos \theta}{r^3 r_0^3}$$

$$\frac{d\theta}{\cos^3 \theta} = A \frac{\cos^3 \theta_0}{r_0^3} \sin(\Omega t) dt$$

$$\frac{2 \cos^2 \theta}{r_0^3 \Omega} [1 - \cos(\Omega t)] = \theta - \theta_0 + \frac{\sin(\theta) - \sin(\theta_0)}{2}$$

$$\textcircled{6} \quad t=0: \theta=0, r_0=\lambda \quad (\vec{x}=\vec{x}_0(\lambda))$$

$$\frac{r}{\cos \theta} = \lambda, \frac{A}{\Omega} [1 - \cos(\Omega t)] = \frac{\lambda^3}{2} \left(\theta + \frac{\sin 2\theta}{2} \right)$$

$$\rightarrow \text{ELIMINATING } \lambda \Rightarrow \frac{r^3}{2 \cos^3 \theta} \left(\theta + \frac{\sin 2\theta}{2} \right) = \frac{A}{\Omega} [1 + \cos(\Omega t)]$$